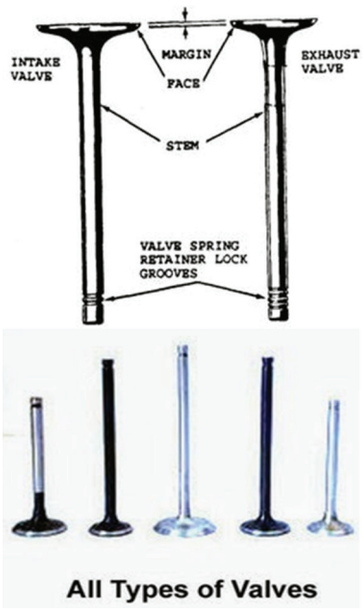


Figure 5.3.13(a) Valves

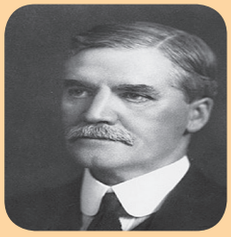
the rocker arm is made in contact with the valve stem. Rocker arm on the rocker shaft is placed in such a way that it could easily moved up and down. Valves are placed in the valve guides on the cylinder head. Valves are seated properly with the help of valve spring and spring lock.

Operation

When the engine runs, the crank shaft rotates. As the crank shaft is connected with the cam shaft with the help of the timing gear and chain arrangement the cam shaft rotates now. As the cam shaft rotates, the cam present on the cam shaft also rotates. Once the cam rotates, the tappet (placed on the cam) starts to move up and down depends on the rotation of the cam. Due to this action, the push rod which is in contact with the tappet starts to move up and down. This movement of the push rod lifts one end of the rocker arm upwards. As the rocker arm is fitted with rocker shaft, the other end of the rocker arm moves down towards valve stem due to the lift of the arm end at the push rod side. Due to this valve moves downwards and opens the port. By repeating the cam shaft rotation, tappet, push rod and rocker arm, both inlet and exhaust valves open and close for inducting the charge and expelling the hot gases.



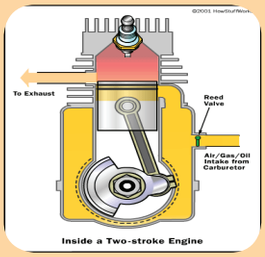
TWO STROKE ENGINE



Sir Dugald Clerk:

Sir Dugald C lerk (1854, Glasgow – 1932, Ewhurst, S urrey) was a Scottish engineer who designed the world' s first successful two-stroke engine in 1878 and patented it in England in 1881.

He was a graduate of Anderson's University in Glasgow (now the University of S trathclyde), and Yorkshire College, Leeds (now the University of Leeds). He formed the intellectual property firm with George Croydon Marks, called Marks & C lerk. He was knighted on 24 August 1917.



Inside a Two-stroke Engine

Dugald C lerk was born in Glasgow on 31 March 1854, the son of Donald Clerk a machinist and his wife, Martha Symington. He was privately tutored thenapprenticed to the firm of Messrs H O Robinson & Co in Glasgow.

From 1871 to 1876 he went to Anderson College in Glasgow studying engineering then to the Yorkshire College of Science in Leeds. In the First World War he was Director of Engineering Research for the Admiralty.

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